THIRD YEAR | Smart Student Marketplace

Level: Advanced | Suggested build time: 3.5–4 hours

Scenario

Students compare second-hand books, bicycles and gadgets but struggle to judge fair prices. Build a lightweight marketplace that supports listings and provides an explainable, budget-aware recommendation.

Input data

ID	Item	Price ₹	Age (years)	Condition	Original ₹
P01	Laptop A	22000	2	Good	45000
P02	Laptop B	18000	3	Average	42000
P03	Laptop C	25000	1	Excellent	48000
P04	Laptop D	20000	2	Good	40000
P05	Laptop E	15000	4	Average	38000

Buyer budget for the example: ₹23,000. Organizers will provide 20–30 listings in CSV/JSON with category, title, price, age, condition and original price. Compare prices only within reasonably similar product categories.

Mandatory tasks

- Load and display listings; allow adding or editing a listing locally.
- Search by name and filter by category and maximum budget.
- Calculate at least one price indicator (for example, price/original-price ratio).
- Implement a documented ranking formula using at least price, age and condition.
- Display the top three eligible products and a reason for each recommendation.

Analytical challenges

- Flag unusually priced listings using a clear within-category rule (for example, deviation from category median); explain why a flag is not proof of fraud.
- Demonstrate how recommendations change when a buyer prioritizes affordability versus condition.
- Handle missing values, identical scores and categories with too few comparable items.

Expected output

Functional marketplace prototype plus a ranked recommendation view showing input factors, calculated score and explanation. Include a short note describing the formula and any assumptions.

Constraints

No real payment or seller authentication. No machine-learning training required. Use only information available in the supplied dataset; do not assume that all products have equivalent specifications.

Evaluation criteria (100 points)

Criterion	Points
Marketplace core features	25
Price analysis and anomaly rule	25
Ranking algorithm	25
Edge-case handling and reasoning	15
Usability and explanation	10